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Eco-friendly perovskite/CZTSSe tandem cell exceeding 28% efficiency through current matching and bandgap optimization: a numerical investigation

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Abstract Tandem solar cells are capable of dividing the solar spectrum over multiple sub-cells with distinct bandgaps, leading to enhanced conversion efficiencies compared to single-junction solar cells. In this paper, we investigate a new eco-friendly tandem solar cell design utilizing lead-free perovskite (LFP)/CZTSSe that can achieve very high efficiency levels is proposed and numerically investigated. The proposed design consists of a LFP-based top cell with wide bandgap of 1.8 eV, and a Cd-free CZTSSe-based bottom cell with graded bandgap profile. Numerical models based on 1-D SCAPS simulator are developed to investigate the photovoltaic performances of the proposed LFP/CZTSSe tandem solar cell. The influence of sub-cell parameters such as layers thickness, unabsorbed spectrum by top cell and GBG paradigm is investigated. The two cells have been improved, leading to an increase in the tandem efficiency and a rise in the matching current. The obtained numerical results show that the highest efficiency value of 28.4% was obtained with top and bottom cell thicknesses of 950 nm and 1 μm, respectively. Our study introduces a novel approach to improving the efficiency of tandem solar cells at an affordable cost. The proposed design methodology resolves optical and electrical losses and toxicity issues and also offers better efficiency.

1 Introduction

The increasing need for energy has created significant environmental concerns regarding the use of fossil fuels [1]. As a result, it has become essential to develop renewable energy sources that are highly efficient, cost-effective, and eco-friendly [1–3]. In this context, photovoltaic (PV) technology is conceded as a potential alternative for clean energy generation. Si-PV is currently the leading player in the PV industry, reaching an efficiency record of 26.6%, which is getting closer to the Schokley-Queisser limit of single-junction solar cells 33% [2–6]. The only way to exceed this limit is by combining two or more individual solar cells with dissimilar bandgaps. This is known as multi-junction technology that allows the absorption of different ranges of wavelength bands, while maintaining reduced thermalization losses [7–10]. Initially, the upper limit for the theoretical efficiency of double-junction tandem solar cells was approximately 42%, but recent theoretical calculations have extended this limit to 46% [11, 12]. In addition, the elaboration of low cost solar cells required the development new design approaches based on thin-film technology using non-toxic and eco-friendly materials such as lead-free perovskites, CIGS, SnS and CZTS [7, 13, 14].

Perovskite solar cells with an efficiency record of 25.7% have recently been introduced as a promising option for developing high-performance low-cost solar cells [15]. These cells use readily available materials and can be produced using simple and low-cost techniques. Perovskite materials are highly promising as wide bandgap top cells or narrow bandgap bottom cells in tandem devices [16–21], due to their bandgap tenability between 1.2 and 2.2 eV. This latter property makes perovskite material as one of the top candidates for developing tandem solar cells capable of capturing a broad spectral range. Several tandem configurations based on perovskite top cell and Si, CIGS, or CZTS bottom cells, have shown the potential to attain efficiencies surpassing 32.5% [22–24]. However, the advancement of stable and highly efficient perovskite-based tandem solar cells is impeded by reliability issues such as toxicity and degradation-related aging effects [19–25]. To address these concerns, scientists have focused their research efforts on lead-free perovskites to eliminate the toxicity of lead. Transition metals like Sn and Ge have been utilized as substitutes for lead due to their comparable electronic structures and their capacity to form new lead-free perovskites. CZTS p-type chalcogenide semiconductor is gaining a great deal of attention for thin-film photovoltaic due to its abundance in the earth, low cost, non-toxic nature, high absorption coefficient (greater than 10^4 cm^{-1}), tunable bandgap and unique optical characteristics [26–28]. More importantly, its inherent advantages make it a potential solution for balancing high-efficiency and low fabrication cost in solar

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cell technology [28–32]. However, the recorded values (12.6%) are still lower than that offered by the CIGS thin film solar cell counterpart (23.4%) [26–29]. This can be attributed to several limitations, such as short minority carrier lifetime and high series resistance caused by the formation of MoS_2 at the CZTS/Mo interface.

This work involves the use of a top sub-cell that utilized a lead-free perovskite (LFP) material ($Cs_2AgBi_{0.75}Sb_{0.25}Br_6$) with a bandgap of 1.8 eV. For the bottom sub-cell, we used a CZTSSe material with a bandgap of 1.097 eV. Accurate numerical models using SCAPS-1D simulator are developed to analyze the performance of the proposed tandem solar cell structure. This study focuses on exploring the effect of band alignment on the photovoltaic performance of both sub-cells, by utilizing appropriate carrier transport layers. Additionally, the research also examines the influence of bandgap engineering on the device's overall performance by applying the graded bandgap (GBG) aspect to the absorber layer in the bottom sub-cell. Moreover, the inclusion of a non-toxic buffer layer offers the potential to improve the alignment of junction bands and enhance conversion efficiency. It is found that the optimized tandem design offers improved efficiency, making it a viable option for developing low-cost, reliable, and eco-friendly tandem solar cells.

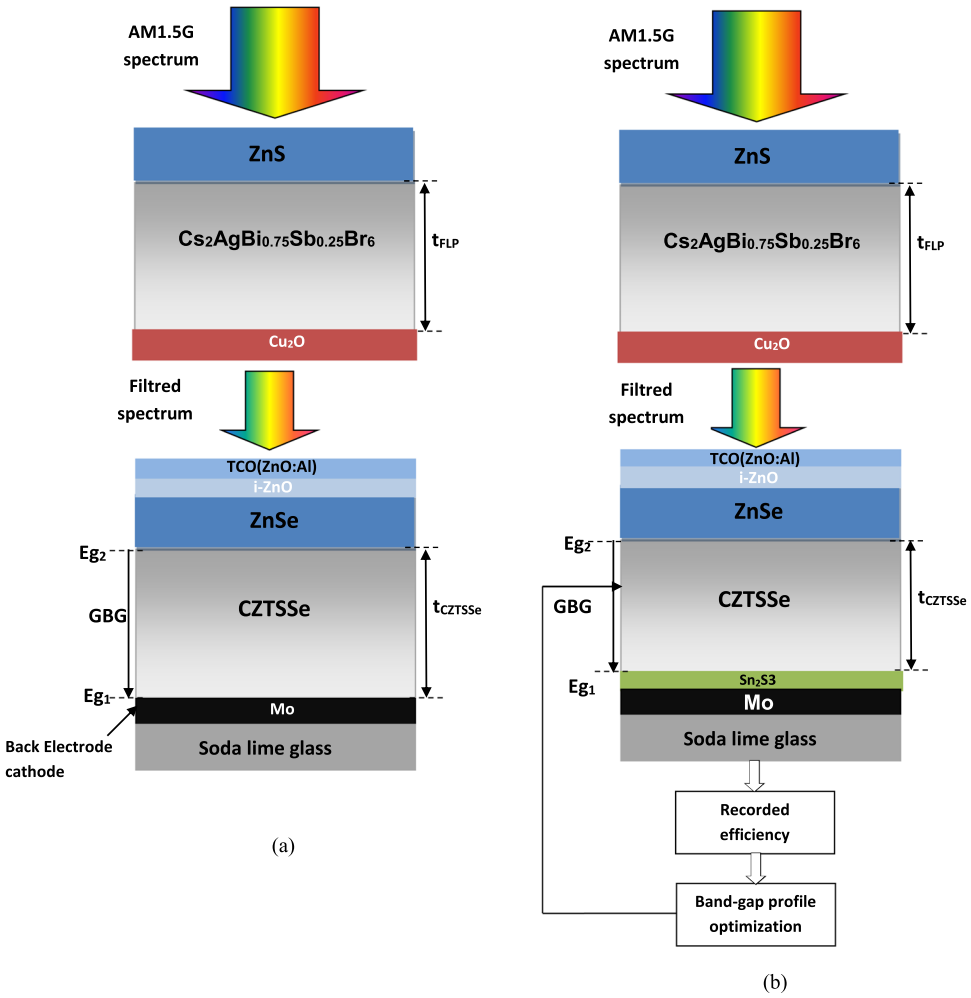
2 Tandem cell architecture and simulation methodology

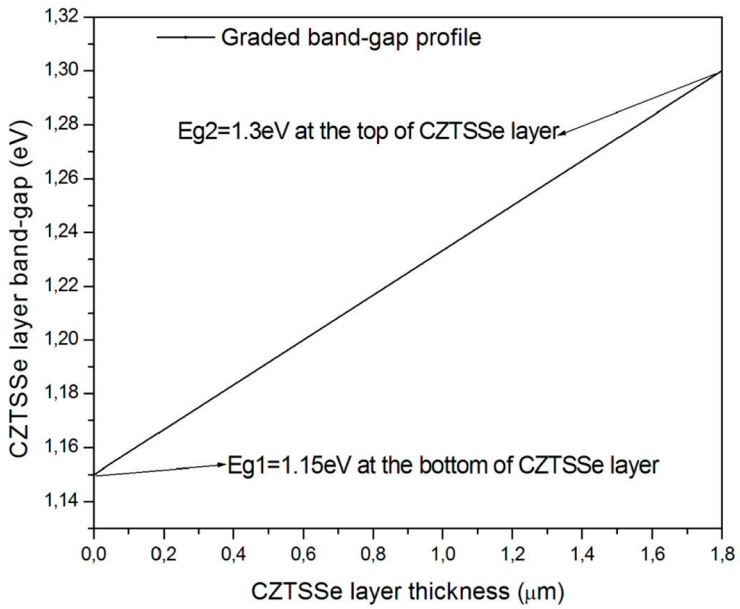
2.1 Device structure and materials

Tandem solar cell configurations can achieve higher efficiencies compared to single-junction cells because their capability for avoiding thermalization effects. In the study, LFP/CZTS tandem architecture is proposed to harvest the solar spectrum. The top cell, made of a high bandgap lead-free perovskite material, generates photocurrent from the short-wavelength region of sunlight, while the longer-wavelength photons are transmitted to a bottom solar cell based on kesterite earth abundant material. The diagram in Fig. 1a illustrates the proposed Top sub-cell LFP configuration of ZnS and copper oxide (Cu_2O) as electron and hole transport layers, respectively. The LFP $Cs_2AgBi_{0.75}Sb_{0.25}Br_6$ material was employed as the absorber layer [33]. The choice of such double perovskite absorber is mainly related to its interesting benefits including wide band gap tenability, enhanced optoelectronic properties and environment-friendly, which make it a potential candidate for developing high-performance cost effective and efficient photovoltaic devices [35, 38]. To achieve high efficiency, the electron and hole transport films will be carefully switched later. The proposed bottom sub-cell structure comprises soda lime glass (SLG) substrate, Mo back contact, CZTSSe absorber layer, ZnSe buffer layer replacing the CdS toxic buffer layer, intrinsic ZnO layer, Al-doped ZnO (AZO) layer acting as TCO, and a 500 nm-thick Al collection grid [34]. The parameters t_{LFP} and t_{CZTSSe} are, respectively, LFP absorber and CZTSSe thicknesses. Besides, the proposed structure consists of a graded band gap CZTS absorber with $E_g(y)$ linear profile as it is shown in Fig. 1c. In this context, Eg_1 and Eg_2 denote the limits bandgap values of the CZTSSe material. Alternatively, the second alteration in design comprises modifying the absorber layer of the bottom cell by implementing an optimized graded bandgap profile of the CZTSSe material and introducing a back surface field (BSF) between the absorber and the bottom electrode cathode Mo. This is done with the intention of achieving two objectives simultaneously, which are the optimization of the conduction band offset (CBO) and the minimization of bulk recombination, which can be further decreased by implementing a BSF, as shown in Fig. 1b.

2.2 Concept of the device fabrication steps

The investigated CZTSSe bottom cell structure consists of soda lime glass with sputtered molybdenum as back contact, graded band gap CZTSSe absorber, ZnSe buffer and AZO buffer layer. To elaborate CZTSSe alloy, Cu, Sn, and Zn layers can be sputtered with a stacking order of Cu/Sn/Zn/Mo on the substrate. Thereafter, annealing in a H_2Se Selenium (H_2Se) atmosphere allows the formation of $Cu_2ZnSnSe_4$ material. A similar supply of H_2S is also useful for depositing Cu_2ZnSnS_4 material [36, 37]. In addition, graded band gap CZTSSe can be elaborated by modifying the standard sequential process [37] or by controlling the sulfo-selenization process, where the variation of SeS_2/Se weight during the annealing process enables grading the [S]/([S] + [Se]) ratio of the CZTSSe [34, 38]. Afterward, ZnSe buffer and Al-doped ZnO (AZO) TCO films can be grown by sputtering technique. Basically, an interconnecting layer (ICL) is important for the realization of reliable monolithically integrated tandem solar cell [39]. In this context, ITO layer acting as (ICL) can be developed on the CZTSSe bottom cell. In addition, the smoothness ICL is equally important because of ultralow thickness of sub-layers of the top perovskite cell, inducing sensitivity to substrate roughness [39]. To address this issue, chemical mechanical polishing (CMP) process can be used to smooth out the ICL surface. The Cu_2O film acting as HTL of the top solar cell can be developed using several methods such as sputtering [40], electrodeposition [41], and metalorganic chemical vapor deposition (MOCVD) [42]. Besides, $Cs_2AgBiSbBr_6$ lead-free double perovskite absorber layer can be synthesized by solution processing route, where CsBr, AgBr, $SbBr_3$ and $BiBr_3$ precursors can be mixed according to desired stoichiometric ratio perovskite material [43, 44]. ZnS buffer layer can be grown using chemical-bath-deposition (CBD) technique [45]. Finally, top Ag electrodes can be coated by thermal evaporation technique.





(c)

Fig. 1 **a** Cross-sectional view of the investigated LFP/CZTSSe tandem solar cell with a random graded bandgap aspect. **b** Cross-sectional view of the optimized LFP/CZTSSe tandem solar cell. **c** Band gap grading profile of the CZTSSe absorber layer

Table 1 Set of the bottom sub-cell parameters adopted for the simulations

Parameters	Sn_2S_3 (BSF)	CZTSSe	ZnSe	i-ZnO	ZnO-Al
Thickness (μm)	0.010	1	0.05	0.05	0.300
Bandgap Eg(eV)	1.09	1.097	2.9	3.3	3.3
Electron affinity χ (eV)	4.26	4.0	4.02	4.4	4.4
Dielectric permittivity	12.5	13.6	10	9	9.0
CB effective density of states (cm^{-3})	1×10^{19}	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}	2.2×10^{18}
VB effective density of states (cm^{-3})	1×10^{19}	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}	1.8×10^{19}
Electron mobility (cm^2/V.s)	25	100	25	100	100
Hole mobility (cm^2/V.s)	100	25	100	25	25
Donor concentration N_D (cm^{-3})	0	0	1×10^{18}	1×10^{18}	1×10^{18}
Acceptor concentration N_A (cm^{-3})	3×10^{18}	1×10^{15}	0	1×10^{18}	0

Table 2 Set of the top sub-cell parameters adopted for the simulations

Parameters	Cu_2O (HTL)	LFP (absorber)	ZnS
Thickness (μm)	0.040	0.4	0.04
Bandgap Eg(eV)	2.17	1.8	2.26
Electron affinity χ (eV)	3.2	3.58	3.5
Dielectric permittivity	7.11	6.5	9.67
CB effective density of states (cm^{-3})	2×10^{17}	2.2×10^{18}	3.7×10^{18}
VB effective density of states (cm^{-3})	1.1×10^{19}	1.8×10^{19}	1.8×10^{19}
Electron mobility (cm^2/V.s)	20	2	80
Hole mobility (cm^2/V.s)	80	2	330
Donor concentration N_D (cm^{-3})	1×10^{7}	1×10^{13}	1×10^{16}
Acceptor concentration N_A (cm^{-3})	1×10^{18}	1×10^{16}	0

2.3 Numerical modeling

To accurately simulate the suggested tandem structure, one could represent the examined LFP/kesterite structure as a series combination of two distinct single-junction solar cells having different bandgap energies (one made of LFP and the other made of CZTSSe material). In this context, the top layer of the solar cell is exposed to the AM1.5G standard spectrum while the bottom layer is exposed to the remaining unabsorbed spectrum. The unabsorbed or the transmitted spectrum simulates the optical behavior of the top layer. To do so, an optical transmission filter is used, which is calculated using the following equation [46, 47].

$$S(\lambda) = S_0(\lambda).\exp(-\alpha_{ITO}.d_{ITO}).\exp(-\alpha_{SnO2}.d_{SnO2}).\exp(-\alpha_{ETL}.d_{ETL}).\exp\left(\alpha_{perov}.d_{perov}\right).\exp(-\alpha_{HTL}.d_{HTL}) \quad (1)$$

where α is the absorption coefficient of each material, $S_0(\lambda)$ is the global AM1.5G spectrum, and d represents the thickness of deferent solar cell layers. The basic characteristics of the structures, which were obtained from previously published literature [33, 48–51] are provided in Tables 1 and 2.

To confirm the accuracy of the developed model, we compared the J-V characteristics obtained through simulation of the conventional LFP and CZTSSe-based solar cells (as shown in Fig. 2) with the experimental data provided in [33, 34]. This figure indicates a close match between the simulated and experimental results, demonstrating the accuracy of our numerical modeling methodology used for analyzing the electrical and optical properties of the proposed tandem cell structure.

This study uses a software tool known as SCAPS (solar cell capacitance simulator), which is adopted to simulate the electronic characteristics of tandem solar cells made from perovskite and kesterite. SCAPS is a software that utilizes important equations to simulate the capture of sunlight, as well as the generation, transport, and extraction of electron/hole pairs [34]. This adaptable tool enables the entry of a broad range of unique materials parameters for the production of a specific solar device. The outputs of simulations are then used to evaluate their impact on the device's performance and obtain crucial insights into fundamental solar characteristics like materials defect density, recombination, location and level, bandgap alignment between device layers, and more. Overall, the use of SCAPS allows answering various fundamental questions related to materials science and device physics [52, 53]. The basis of the theoretical calculation for semiconductors consists of three fundamental equations, which are Poisson's equation, the carrier continuity equation, and the drift–diffusion equation,

$$\frac{\mathrm{d}p_n}{\mathrm{d}t} = G_p - \frac{p_n - p_{n0}}{\tau_p} - p_n u_p \frac{\mathrm{d}E}{\mathrm{d}x} - u_p E \frac{\mathrm{d}p_n}{d_x} + D_p \frac{d^2 p_n}{\mathrm{d}x^2} \quad (2)$$

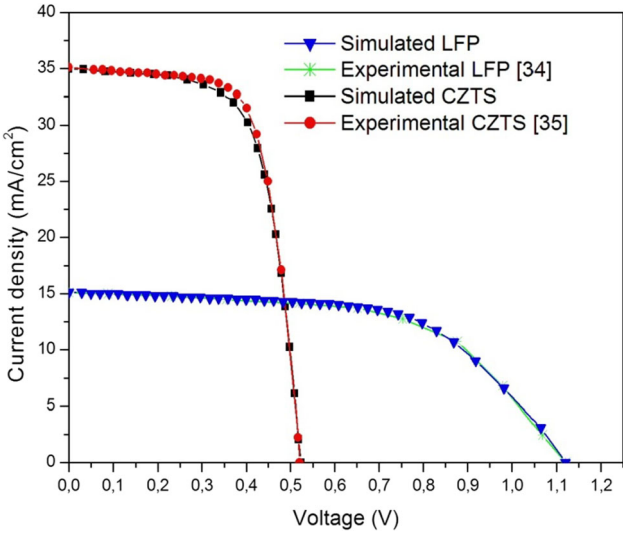


Fig. 2 J–V characteristics of experimental data and simulation results associated with both the conventional LFP and CZTS solar cells

$$\frac{\mathrm{d}n_p}{\mathrm{d}t} = G_n - \frac{n_p - n_{p0}}{\tau_n} - n_p u_n \frac{\mathrm{d}E}{\mathrm{d}x} - u_n E \frac{\mathrm{d}n_p}{d_x} + D_n \frac{d^2 n_p}{\mathrm{d}x^2} \tag{3}$$

$$\frac{d}{\mathrm{d}x}\left(\varepsilon(x)\frac{d\phi}{dx}\right) = q\left[p(x) - n(x) + N_{d^+}(x) - N_{a^-}(x) + p_t(x) - n_t(x)\right] \tag{4}$$

where ε stands for the dielectric permittivity, q represents the electron charge, G represents the generation rate, D designates the diffusion coefficient, φ represents the electrostatic potential, E refers to the electric field, $p(x)$ and $n(x)$ represent the free holes and electrons, and $p_t(x)$ and $n_t(x)$ denote the trapped holes and electrons. N_{d+} and N_{a-} represent the donor and acceptor ionized doping concentrations, respectively, and x represents the thickness.

3 Results and discussion

Perovskite is considered the best choice for creating high-performance and cost-effective tandem solar cells. The ETL and HTL layers are crucial for determining the performance of perovskite solar cells. To improve the photovoltaic efficiency of the free-lead PSC design, various ETLs and HTLs were explored. It was discovered that using cuprous oxide Cu_2O as the HTL was the most effective way to enhance the PSC's efficiency [54]. This is due to the small activation energy of interfacial recombination, which is the difference between the band gap of the absorber layer and the valence band offset (*VBO*) [55]. The activation energy for carrier recombination can be expressed as:

$$\mathrm{VBO} = \mathrm{Ev}_{(\mathrm{HTL})} - \mathrm{Ev}_{(\mathrm{Absorber\ Layer})} \tag{5}$$

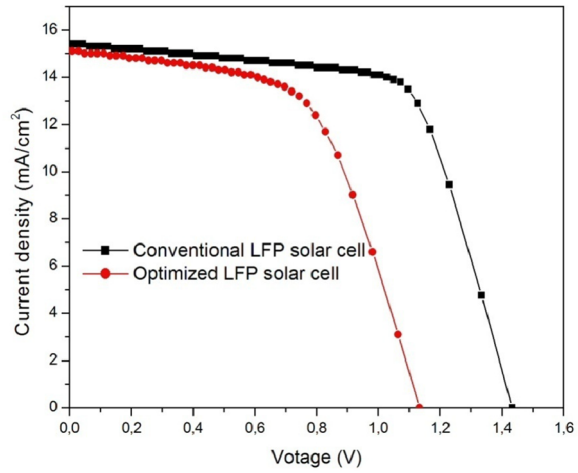
where $Ev_{(\mathrm{HTL})}$, and $Ev_{(\mathrm{Absorber\ Layer})}$ denote the energy valance band minimum of the HTL and the Absorber Layer, respectively. In the same way, ETL and perovskite absorber interface has a significant impact on performance of the device, if the band alignment between the perovskite absorber and the ETL is not appropriate, it can affect the shunt and series resistance, as well as recombination resistance of the device. To address these issues, it is important to carefully select materials with high electron mobility and proper band alignment for the ETL/perovskite interface to provide an excellent conduction band offset (CBO). This will improve electron injection and hole blocking, leading to higher current density J_{sc} and better overall device performance [56].

$$\mathrm{CBO} = \mathrm{Ec}_{(\mathrm{ETL})} - \mathrm{Ec}_{(\mathrm{Absorber\ Layer})} \tag{6}$$

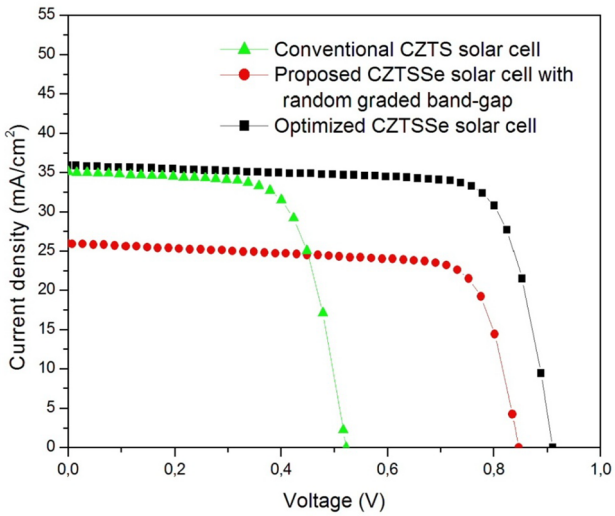
which indicates the difference between CB level of ETL and that of the absorber layer.

It was found that the best ETL choice is the ZnS with its high electron affinity, which enables efficient electron extraction from the perovskite layer. Additionally, ZnS has a wide bandgap, which minimizes energy losses due to recombination and ensures high open-circuit voltage in the device [57]. In this perspective, Fig. 3a displays the J–V curve linked to the optimized LFP solar cell compared to the conventional structure. The figure illustrates that the proposed design has better photovoltaic characteristics than the conventional design, with a higher output voltage of 1.43 V, an improved efficiency of 14.83%, and a FF of 67.17%. These enhancements are due to the selection of ETL/HTL, which possesses improved band alignment, reducing interface recombination effects and allowing for a reduction in V_{OC} deficit.

Selecting narrow bandgap materials for the bottom cells is also crucial for achieving superior output voltage and high efficiency. Under standard AM1.5G illumination, in Fig. 3b the J–V characteristics of the optimized design were compared to that of both the



(a)



(b)

Fig. 3 J–V characteristics comparison between: **a** the optimized LFP and the conventional structures **b** the optimized design and that of both the proposed structure with a random bandgap profile and the conventional structures counterparts

proposed structure with a random bandgap profile and the conventional structures counterparts. The main objective for utilizing a CZTSSe absorber layer with a graded bandgap profile is to improve the performance of kesterite solar cells, by enhancing the absorption characteristics across the complete solar spectrum. Besides, the usage of the graded bandgap approach creates a consistent potential gradient in the CZTSSe absorber layer, creating an extra electric field that facilitates efficient separation of electron–hole pairs and boosts the carrier's lifetime. For further more enhancement, a BSF layer of Sn_2S_3 was introduced between CZTSSe and Mo layers, to create a barrier for electrons, which can reduce the carrier recombination rate in the backside of the device. To prevent toxicity, the buffer layer used was the Zinc Selenide (ZnSe), a semiconductor compound with a direct band gap [58], which considered to be one of the most favorable options among various other possible substitute materials [59, 60]. Figure 3b provides clear evidence that the suggested designs demonstrate better performance in terms of photocurrent (approximately 35.92 mA/cm^2) open-circuit voltage (around 0.91 V), and a fill factor of 77.44 when compared to the standard version. Furthermore, the proposed design leads to higher conversion efficiency (approximately 24.32%) compared to the conventional design (12.30%). The photovoltaic parameters of the devices are summarized in Table.3 alongside comparison against experimentally fabricated perovskite and CZTS solar cells with wide bandgap and graded bandgap of top and bottom cells, respectively.

3.1 Current matching

The band gap has an impact on the theoretical yield of a single-junction solar cell. A large band gap can be used for a high-voltage, but this will give a low current. Conversely, if the band gap is reduced to absorb most of the spectrum, the voltage of the cell decreases. However, a multi-junction cell can efficiently collect low-energy photons while still achieving a high-voltage cell [61]. In this perspective, the top cell is exposed to an AM1.5 spectrum, and the transmitted spectrum $S(\lambda)$ is determined by using the

Table 3 Performance comparison between conventional and optimized structures

Devices	V_{OC}(V)	J_{SC}(mA/cm^2)	FF(%)	PCE(%)
Experimental LFP	1.12	15.1	58.5	10.08
Optimized LFP	1.43	15.4	67.17	14.83
Experimental CZTS	0.52	34.98	67.20	12.30
Proposed GBG CZTSSe	0.84	25.97	75.41	16.56
Optimized GBG CZTSSe	0.91	34.92	77.44	24.32

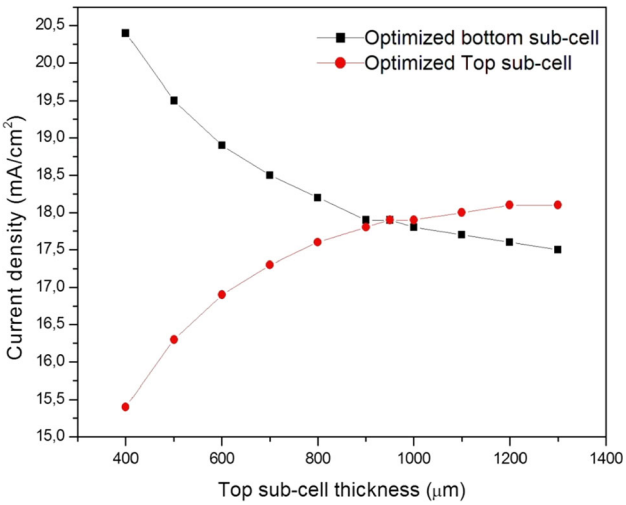
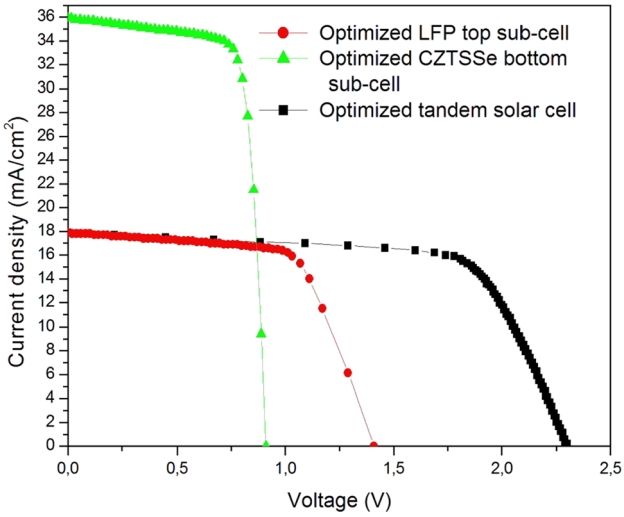


Fig. 4 The current matching condition after optimization

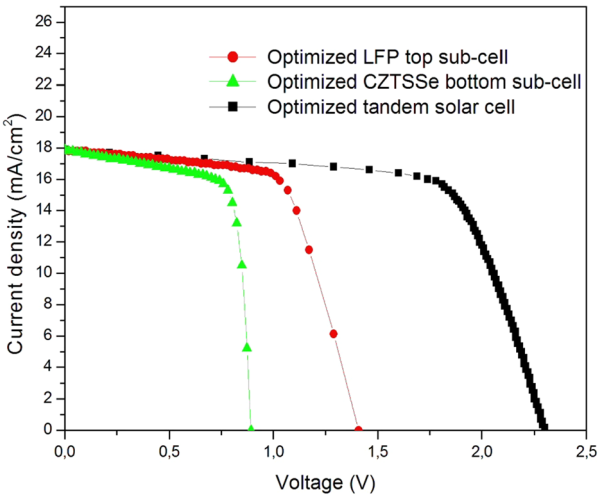
absorption coefficient and thickness of all layers in the top cell, as indicated in Eq. (1). Firstly, the top cell is modeled by changing the absorber layer thickness from 400 to 1300 nm, while keeping other layers' thickness constant. As the absorber layer thickness increases in the top cell, the power of the filtered transmitted spectrum decreases, particularly at shorter wavelengths, which is due to greater absorption in the top cell with a thicker absorber layer. This filtered spectrum is then used to illuminate the bottom cell. According to reference [33], increasing the thickness of the top cell results in a decrease in the transmitted photocurrent due to the greater absorbed photocurrent.

The two-terminal tandem solar cell behaves like two diodes connected in series. As a result of this configuration, an equal amount of current must pass through each cell at all times. Furthermore, the total voltage across the tandem device is the sum of the voltage across each individual cell. Thus, in a tandem device, the cell that has a lower short current acts as the limiting cell for the overall J_{SC}, while the total open-circuit voltage is the sum of the individual V_{OC} values of each cell [62]. Consequently, to ensure an equivalent J_{SC} value in tandem devices, the thickness of both the top and bottom cells are optimized and the same tunnel recombination junctions are utilized in our modeling approach [63, 64]. Furthermore, if the thickness of the top cell is increased beyond the optimized value, it leads to increased parasitic absorption in the top cell and reduced optical coupling in the bottom cell, resulting in a decreased overall J_{SC} value for the tandem device. Similarly, if the thickness of the top cell is decreased below the optimized value, it results in reduced absorption in the top cell and the same consequence of decreased J_{SC}. Taking these factors into account, the conditions for current matching are determined in order to achieve equal J_{SC} values in both the top and bottom cells. This is accomplished by analyzing the computed filtered spectrum with varying thicknesses of the absorber layer in the top cell. In order to address the variation in thickness of the top cell, ten different filtered spectra are generated (corresponding to top cell thicknesses ranging from 400 to 1300 nm) and used to illuminate the bottom cell. The photovoltaic parameters of the bottom cell are then assessed based on these spectra. Using the J_{SC} values of the bottom cell from the filtered spectrum, the current matching condition for the tandem device was determined and illustrated in Fig. 4. The optimal current matching condition was achieved when the top cell had a thickness of 950 nm and the bottom cell had a thickness of 1000 nm. As shown in Fig. 4, the J_{SC} values of the top and bottom sub-cells were 17.9 mA/cm^2 and 17.8 mA/cm^2, respectively.

The J-V characteristics of the LFP/CZTSSe tandem solar cell can be simulated using the current matching condition determined above. The separately measured J–V curves of both the top and bottom cell at equal current are added together in series, meaning that their voltages are summed at equal current to obtain the J–V characteristics of the tandem device [46, 63]. In Fig. 5a, The J–V curve of the tandem device is presented along with the J–V curve of the standalone top and bottom cells, which are illuminated with AM1.5sun spectrum. It can be observed from this figure that the top cell (LFP) is the cell that restricts the current, having a higher V_{OC} (as a result of its high bandgap perovskite). In Fig. 5b, the J–V plot of the top sub-cell illuminated with AM1.5sun spectrum, the bottom sub-cell illuminated with the filtered spectrum, and the tandem solar cell are shown. It is clear from the plot that the V_{OC} of the tandem solar cell, which is 2.31 V, is the sum of the top and bottom solar cells' V_{OC}, which are 1.41 V and 0.90 V,



(a)



(b)

Fig. 5 **a** J–V curve of the standalone top, standalone bottom and tandem cell. **b** J–V curve of top cell, bottom cell after fed with filtered spectrum and tandem cell

Table 4 Photovoltaic performance of top LFP sub-cell, bottom CZTSSe sub-cell with filtered spectrum and the tandem solar cell

Devices	V_{OC}(V)	J_{SC}(mA/cm^2)	FF(%)	PCE(%)
Top LFP SC	1.41	17.9	65.07	16.45
Bottom CZTSSe SC with filtered spectrum	0.9	17.85	74.99	11.94
Tandem SC	2.31	17.9	69	28.42

respectively. Furthermore, the photovoltaic parameters of the cells are presented in Table.4, where we found that the tandem solar cell has a J_{SC} of 17.9 mA/cm^2, FF of 69%, and PCE of 28.42%.

In order to assess the effectiveness of the proposed approach in enhancing the tandem solar cell performance, the J–V curves for both the conventional structure and the optimized design with graded bandgap profiles were compared in Fig. 6. This figure confirms that a significant improvement was achieved by using the GBG optimizer side in combination with the above optimization techniques, which has resulted in a significant enhancement in the cell performance. So that, there has been an improvement in both short circuit current, open-circuit voltage, and PCE. In Table.5, we have compared the results performance of our work with those of other studies, to evaluate the effectiveness of our optimized tandem cell in comparison to other research studies. This comparison is performed based on the solar cell performance metrics obtained from different tandem designs (Perovskite-CZTSSe [53], Perovskite-Perovskite [65], CZTS-CZTSSe [66], Perovskite-CIGS [67] and CZTS-Perovskite tandem solar cell [68]). The comparison indicates that our optimized design offered high Figures-of-Merit (FoMs) in comparison with that provided by other published studies [53, 65–68]. Moreover, in order to investigate the impact of top sub-cell bandgap profile on the tandem cell performances, we have implemented and optimized the graded bandgap profile of the perovskite absorber material by following the

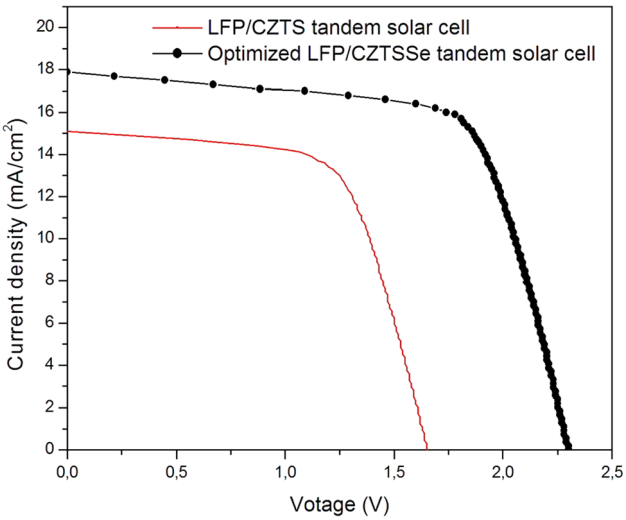


Fig. 6 J–V characteristics of the optimized tandem design compared to the conventional tandem cell

Table 5 Performance comparison between different tandem cell designs and our optimized perovskite/CZTSSe tandem solar cell

Device	V_{OC}(V)	J_{SC}(mA/cm^2)	FF(%)	PCE(%)
Perovskite/perovskite tandem cell [55]	1.92	14.00	78.10	21.00
CZTS/CZTSSe tandem cell [56]	1.10	20.40	81.30	19.25
Perovskite/CIGS tandem cell [57]	1.68	19.17	71.9	24.2
CZTS/perovskite tandem cell [58]	1.18	24.79	88.36	25.95
Perovskite/CZTSSe tandem cell [42]	1.70	14.90	65.30	16.58
This work	2.31	17.90	69	28.42

same steps, which have been previously utilized for the bottom sub-cell (kesterite material); but no improvement has been recorded using double graded bandgap optimization approach. This could be attributed to the range of bandgap grading profile in the LFP cell (from 1.8 to 2.4 eV) [33, 69]. The obtained results make the proposed design methodology a potential alternative approach for developing low-cost and high performance non-toxic tandem thin film solar cells. Moreover, the proposed design approach can be extended by including metaheuristic global optimization techniques and DFT-based simulations to investigate and improve the optoelectronic and degradation properties of the tandem cells. However, new complex computations and modeling frameworks associated with experimental measurements should be developed [3, 70–72].

4 Conclusion

In this paper, we have proposed a new design methodology to improve the thin-film tandem solar cells efficiency, made up of two sub-cells, a lead-free perovskite-based top sub-cell and a CZTSSe-based bottom sub-cell. The numerical analysis aimed to enhance the interfaces of the electron and hole transport layers to improve the band alignment and reduce the recombination effects, which ultimately led to a higher-performing LFP-cell. The performance of CZTSSe sub-cell was improved by introducing a graded bandgap profile, a BSF layer, and a non-toxic buffer layer (ZnSe). The simulation involved exposing the top sub-cell to the AM 1.5G sun spectrum and a filtered spectrum for the bottom sub-cell. The thickness of the two absorber layers was varied to investigate the current matching condition, where the optimized thicknesses of 950 nm and 1000 nm were found for the top and bottom absorber layers, respectively. A current-matched device with a J_{SC} of 17.9 mA/cm^2 was obtained, resulting in PCEs of 16.45% and 11.94% for the top and bottom cells, respectively, and a final tandem structure PCE of 28.42%. Therefore, this work proposed a design framework methodology offering new insights into reducing recombination losses and enhancing absorbance efficiency, which could lead to the development of high-efficiency thin-film tandem solar cells for high-performance non-toxic photovoltaic applications. The study also highlights the importance of operating both cells at the maximum power condition to obtain an efficient tandem cell.

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Data Availability Statement The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no competing interests.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

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